final project

Zhiying(Winnie)Chen

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Contents

0.0.1 My choices

- Gene: GNLY
- Continuous covariate: mat_vitD (vitamin D, ng/mL)
- Categorical covariate 1: mat_BMI (BMI group, derived from a continuous variable)
- Categorical covariate 2: preg_preeclampsia (Normal vs. Preeclampsia)

0.0.2 Step 1. Load the raw data

```
#read.csv with row.names = 1 uses the first column (probe IDs) as row names
covar <- read.csv(file = 'GSE85307_covar-1.csv')
geneNames <- read.csv(file = 'GSE85307_geneNames.csv')
expr <- read.csv(file = 'GSE85307_geneExpression.csv', row.names = 1)

# The first column name of the covariate file can come in with a hidden extra
# character (BOM), so we rename it ourselves
colnames(covar)[1] <- 'subjectID'

# Look at the the structure
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                   : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-
M" ...
##  $ geoID                       : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
##  $ mat_age_yrs                 : chr  "age_years: 31.06639288" "age_years: 27.79466119" "age_years: 3
##  $ preg_preeclampsia           : chr  "pregnancy_condition: Preeclampsia" "pregnancy_condition: Norma
##  $ delivery_gestAge_wks        : chr  "gestation_weeks: 37.42857143" "gestation_weeks: 39.42857143"
##  $ sampleCollection_gestAge_wks: chr  "blood_draw_at_gestation_week: 10" "blood_draw_at_gestation_we
##  $ batch_Microarray            : chr  "microarray_hybridization_date: 03/15/2013" "microarray_hybridi
##  $ mat_BMI                     : chr  "bmi_kg/m2: 37.52407538" "bmi_kg/m2: 20.59144273" "bmi_kg/m2: 1
##  $ mat_vitD                    : chr  "vitamind_25(0h)_ng/ml_whole_blood_baseline: 7.94" "vitamind_25
##  $ mat_asthma                  : chr  "asthma_0=healthy_1=asthmatic: 1" "asthma_0=healthy_1=asthmatic
```

```
dim(expr)
```

```
## [1] 500 157
```

```
head(geneNames)
```

```
##   PROBEID      GENE
## 1 8128043      CNR1
## 2 7962489 PLEKHA8P1
## 3 7920271    S100A4
## 4 7900468      NFYC
## 5 8026496 LINC00661
## 6 7989725     PDCD7
```

0.0.3 Step 2. Clean the covariates

Every covariate value is stored as text with a label, like "age_year: xxx". I remove everything up to and including ":" and then fix the data types.

```
# Loop through columns 3 to 10 (all covariates; columns 1-2 are the IDs)
# '.*: ' means "any characters followed by a colon and a space"
for(i in colnames(covar)[3:10]){
  covar[,i] <- sub(pattern = '.*: ',replacement = '', x = covar[,i])
}
head(covar)
```

```
##   subjectID      geoID mat_age_yrs preg_preeclampsia delivery_gestAge_wks
## 1 6053887-M GSM2264346 31.06639288      Preeclampsia          37.42857143
## 2 3064425-M GSM2264347 27.79466119            Normal          39.42857143
## 3 6094898-M GSM2264348 34.31895962            Normal          38.85714286
## 4 7050617-M GSM2264349 33.42642026            Normal          39.14285714
## 5 7016015-M GSM2264350 28.27926078      Preeclampsia          40.14285714
## 6 7054360-M GSM2264351 23.50171116            Normal          41.14285714
##   sampleCollection_gestAge_wks batch_Microarray     mat_BMI mat_vitD mat_asthma
## 1                           10       03/15/2013 37.52407538     7.94          1
## 2                  17.42857143       03/15/2013 20.59144273     40.3          1
## 3                           13       03/15/2013 20.71118512     40.8          1
## 4                  17.28571429       03/15/2013 31.99358316     52.2          0
## 5                  10.71428571       03/15/2013 27.17734484     22.8          0
## 6                  10.57142857       03/19/2013 22.28684666     44.6          0
```

```
# Numeric variables
covar$mat_age_yrs <- as.numeric(covar$mat_age_yrs)
covar$delivery_gestAge_wks <- as.numeric(covar$delivery_gestAge_wks)
covar$sampleCollection_gestAge_wks <- as.numeric(covar$sampleCollection_gestAge_wks)
covar$mat_BMI <- as.numeric(covar$mat_BMI)
covar$mat_vitD <- as.numeric(covar$mat_vitD)

# Categorical variables
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia,
                                  levels = c('Normal','Preeclampsia'))
covar$mat_asthma <- factor(ifelse(covar$mat_asthma=='1','Asthma','Healthy'))

# Check that the types and values look right
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                   : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-
M" ...
##  $ geoID                       : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
##  $ mat_age_yrs                 : num  31.1 27.8 34.3 33.4 28.3 ...
##  $ preg_preeclampsia           : Factor w/ 2 levels "Normal","Preeclampsia": 2 1 1 1 2 1 1 1 1 1 ...
##  $ delivery_gestAge_wks        : num  37.4 39.4 38.9 39.1 40.1 ...
##  $ sampleCollection_gestAge_wks: num  10 17.4 13 17.3 10.7 ...
##  $ batch_Microarray            : chr  "03/15/2013" "03/15/2013" "03/15/2013" "03/15/2013" ...
##  $ mat_BMI                     : num  37.5 20.6 20.7 32 27.2 ...
##  $ mat_vitD                    : num  7.94 40.3 40.8 52.2 22.8 44.6 26.2 21.8 39.2 42.9 ...
##  $ mat_asthma                  : Factor w/ 2 levels "Asthma","Healthy": 1 1 1 2 2 2 1 1 2 1 ...
```

```
summary(covar)
```

```
##      subjectID         geoID      mat_age_yrs       preg_preeclampsia
##  Length   :157   Length   :157   Min.   :18.43   Normal      :110
##  N.unique :157   N.unique :157   1st Qu.:22.15   Preeclampsia: 47
##  N.blank  :  0   N.blank  :  0   Median :26.57
##  Min.nchar:  9   Min.nchar: 10   Mean   :26.38
##  Max.nchar:  9   Max.nchar: 10   3rd Qu.:30.14
##                                  Max.   :38.76
##  delivery_gestAge_wks sampleCollection_gestAge_wks  batch_Microarray
##  Min.   :26.43        Min.   :10.00                 Length   :157
##  1st Qu.:38.14        1st Qu.:12.14                 N.unique :  8
##  Median :39.29        Median :13.43                 N.blank  :  0
##  Mean   :38.82        Mean   :14.13                 Min.nchar: 10
##  3rd Qu.:40.00        3rd Qu.:16.43                 Max.nchar: 10
##  Max.   :42.00        Max.   :18.86
##     mat_BMI         mat_vitD       mat_asthma
##  Min.   :17.95   Min.   : 5.11   Asthma :63
##  1st Qu.:22.90   1st Qu.:13.40   Healthy:94
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
```

0.0.4 Step 3. Derive a categorical variable from a continuous one

```
# BMI groups: <25, 25-30,  30
# Only 5 women have the BMI < 18.5, too few for their own group, so they are
# combined with the "<25" group
covar$bmi_group <- cut(covar$mat_BMI,
                       breaks = c(0,25,30,Inf),
                       labels = c('<25','25-30',' 30'))

# Check the sample sizes
table(covar$bmi_group)
```

```
##
##   <25 25-30    30
##    53    47    57
```

0.0.5 Step 4. Find my gene and match it to the covariates

The expression columns are in a different order than the rows of the covriate file, so I cannot just paste them together. I sort both by sample ID and then check that they line up.

```
geneOfInterest <- 'GNLY'
# Gene name -> probe ID (the row names of expr are the probe IDs, stored as text)
probeID <- geneNames[which(geneNames$GENE==geneOfInterest),'PROBEID']
probeID <- as.character(probeID)

# Make sure there is exactly one probe and that it is in the expression data
length(probeID)
```

```
## [1] 1
```

```
probeID %in% rownames(expr)
```

```
## [1] TRUE
```

```
# Transpose so samples are rows and pull out the gene's column
exprT <- t(expr)
geneVec <- exprT[,probeID]

# Sort both the covariates and the gene values by sampleID
covar <- covar[order(covar$geoID),]
geneVec <- geneVec[order(names(geneVec))]

# Check every sample must line up(all True)
table(covar$geoID==names(geneVec))
```

```
##
## TRUE
##  157
```

```
# Only add the gene to the data frame
covar$Expression <- as.numeric(geneVec)
plotData <- covar

# Spot check one sample against the raw expression table (should be TRUE)
expr[probeID,plotData$geoID[1]] == plotData$Expression[1]
```

```
## [1] TRUE
```

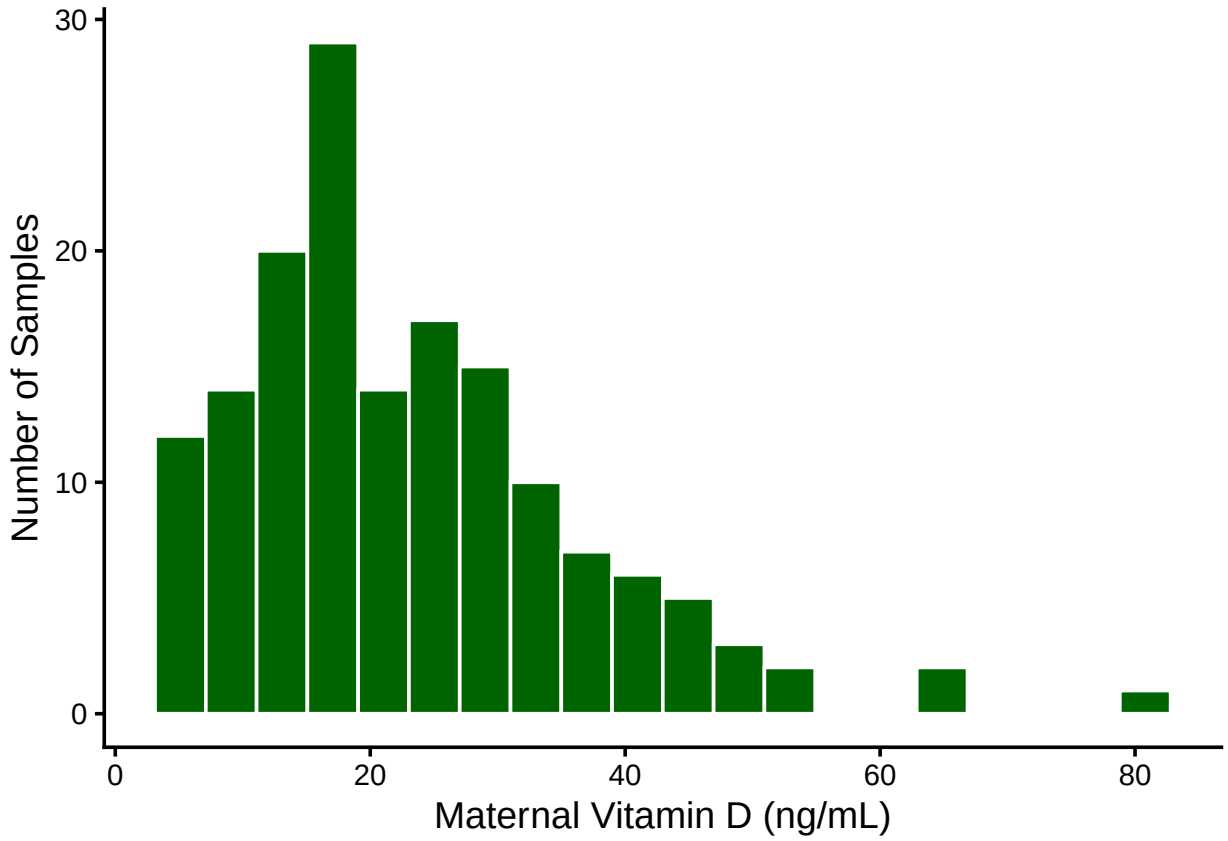
```
# Check the sample sizes per group
table(plotData$bmi_group,plotData$preg_preeclampsia)
```

```
##
##         Normal Preeclampsia
##   <25       40           13
##   25-30     39            8
##    30       31           26
```

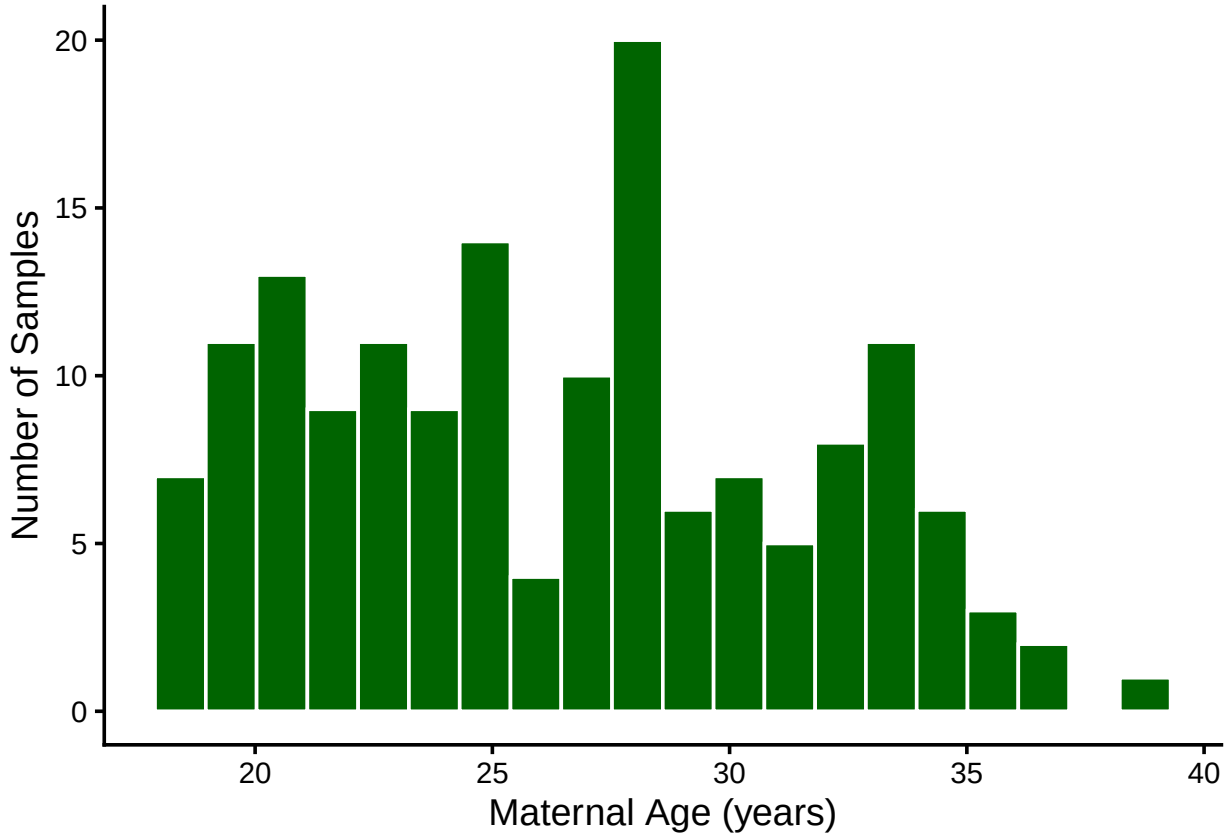
0.0.6 Table 1: summary statistics stratified by preganacy condition

```
# 3 continous variables and their labels
contVars <- c('mat_vitD','mat_age_yrs','delivery_gestAge_wks')
contLabels <- c('Maternal vitamin D (ng/mL)',
                'Maternal age (years)',
                'Gestational age at delivery (weeks)')

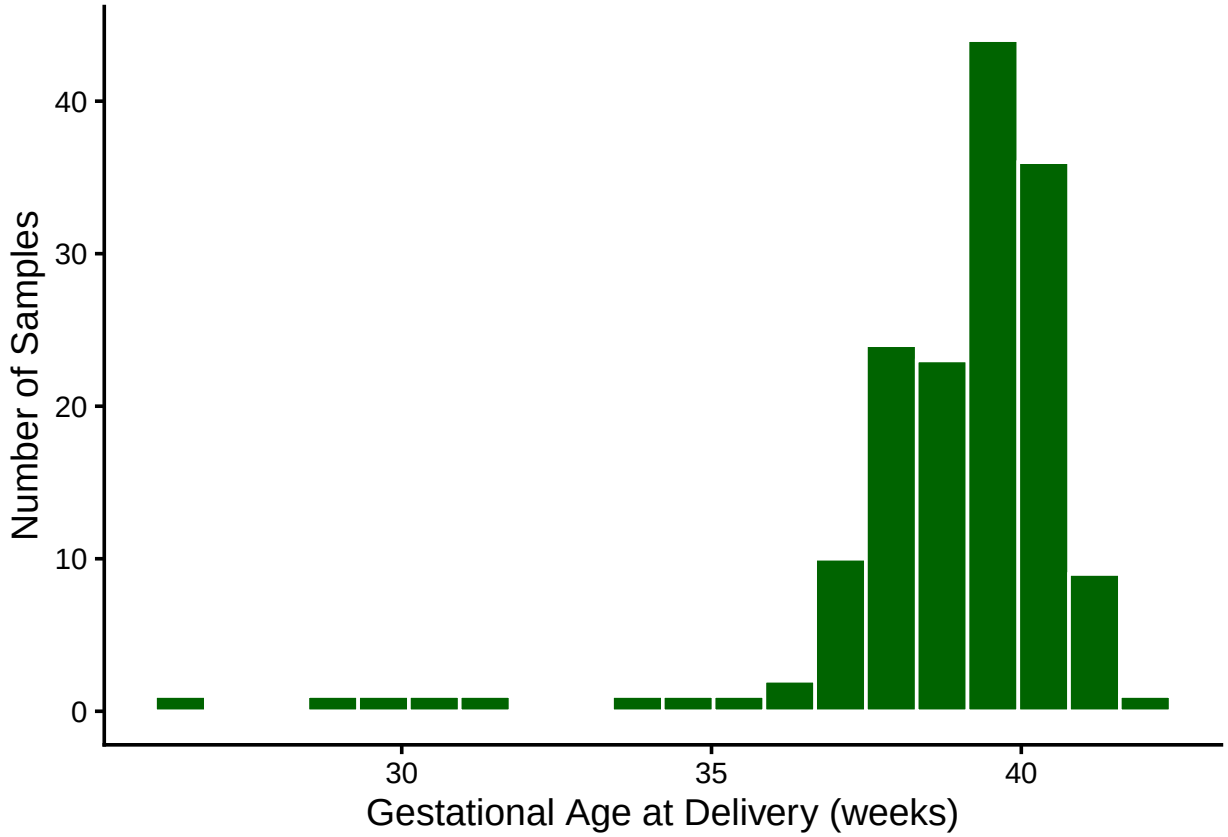
# Look at the histogram of each continous variable
ggplot(covar,aes(x=mat_vitD))+
  geom_histogram(bins=20,fill='darkgreen',color='white')+
  labs(x='Maternal Vitamin D (ng/mL)',y = 'Number of Samples') +
  theme_classic(base_size = 14)
```




```
ggplot(covar,aes(x = mat_age_yrs)) +
  geom_histogram(bins = 20,fill = 'darkgreen',color = 'white') +
  labs(x = 'Maternal Age (years)',y = 'Number of Samples') +
  theme_classic(base_size = 14)
```




```
ggplot(covar,aes(x = delivery_gestAge_wks)) +
  geom_histogram(bins = 20,fill = 'darkgreen',color = 'white') +
  labs(x = 'Gestational Age at Delivery (weeks)',y = 'Number of Samples') +
  theme_classic(base_size = 14)
```




```
# Variables that are not normalization are reported as median [IQR],
# the others as mean(SD). Vitamin D has a long tail of high values and
# gestational age at delivery has a long tail of early deliveries.
nonnormVars <- c('mat_vitD','delivery_gestAge_wks')

# Groups to summarize: all women, then each pregnancy condition
groups <- list('Overall' = rep(T,nrow(covar)),
               'Normal' = covar$preg_preeclampsia == 'Normal',
               'Preeclampsia' = covar$preg_preeclampsia == 'Preeclampsia')

# Start the table with the variable names (and which summary is used)
table1 <- data.frame('Variable' = paste0(contLabels,
                                         ifelse(contVars %in% nonnormVars,
                                                ', median [IQR]',', mean (SD)')))

# Continous rows: loop through the groups, then through the variables
# sprintf keeps 2 decimals
for (g in names(groups)) {
  # Define somewhere to store this column of the table
  colValues <- c()
  for (v in contVars) {
    # Values of this variable in this group
    x <- covar[groups[[g]],v]
    if (v %in% nonnormVars) {
      # Median [IQR]
      colValues[v] <- sprintf('%.2f [%.2f, %.2f]',median(x),
```

```
                              quantile(x,1/4),quantile(x,3/4))
    }
    else {
      # Mean (SD)
      colValues[v] <- sprintf('%.2f (%.2f)',mean(x),sd(x))
    }
  }
  table1[,g] <- colValues
}

# Categorical rows: BMI group, n (% of the column)
bmiRows <- data.frame('Variable' = c('BMI group (kg/m\u00B2), n (%)',
                                     levels(covar$bmi_group)))
for (g in names(groups)) {
  n <- table(covar$bmi_group[groups[[g]]])
    bmiRows[,g] <- c('',paste0(n,' (',sprintf('%.1f',n/sum(n)*100),')'))
}
table1 <- rbind(table1,bmiRows)

# Put the sample size in the column names
colnames(table1) <- c('Variable',
                      paste0('Overall (n = ',sum(groups$Overall),')'),
                      paste0('Normal (n = ',sum(groups$Normal),')'),
                      paste0('Preeclampsia (n = ',sum(groups$Preeclampsia),')'))
table1
```

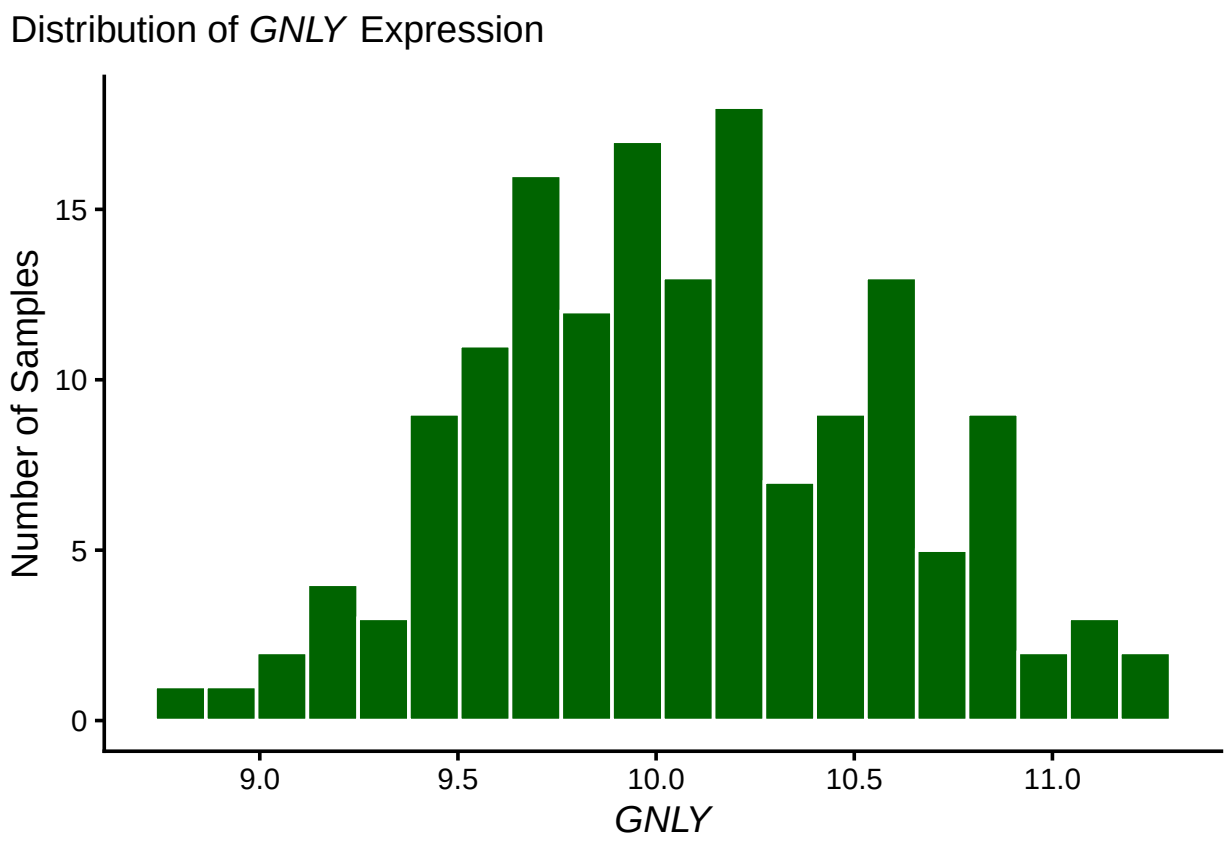
```
##                                           Variable    Overall (n = 157)
## 1          Maternal vitamin D (ng/mL), median [IQR] 20.40 [13.40, 29.50]
## 2                   Maternal age (years), mean (SD)         26.38 (5.04)
## 3 Gestational age at delivery (weeks), median [IQR] 39.29 [38.14, 40.00]
## 4                           BMI group (kg/m²), n (%)
## 5                                               <25            53 (33.8)
## 6                                             25-30            47 (29.9)
## 7                                                30            57 (36.3)
##       Normal (n = 110) Preeclampsia (n = 47)
## 1 21.80 [13.85, 32.08]  17.70 [12.55, 27.40]
## 2         26.63 (5.08)          25.78 (4.95)
## 3 39.29 [38.57, 40.00]  39.14 [36.93, 40.07]
## 4
## 5            40 (36.4)             13 (27.7)
## 6            39 (35.5)              8 (17.0)
## 7            31 (28.2)             26 (55.3)
```

```
write.csv(table1,file = 'final_table1.csv',row.names = F)
```

0.0.7 Final Plot 1. Histogram of gene expression

```
ggplot(plotData,aes(x=Expression)) +
  geom_histogram(bins=20,fill='darkgreen',color='white')+
  labs(title =
         expression(paste('Distribution of ', italic('GNLY'), ' Expression')),
```

```
    x=expression(paste(italic('GNLY')),'Expression'),
    y='Number of Samples')+
  theme_classic(base_size = 14)+
  theme(plot.title = element_text(size = 14),
                    plot.title.position = 'plot')
```



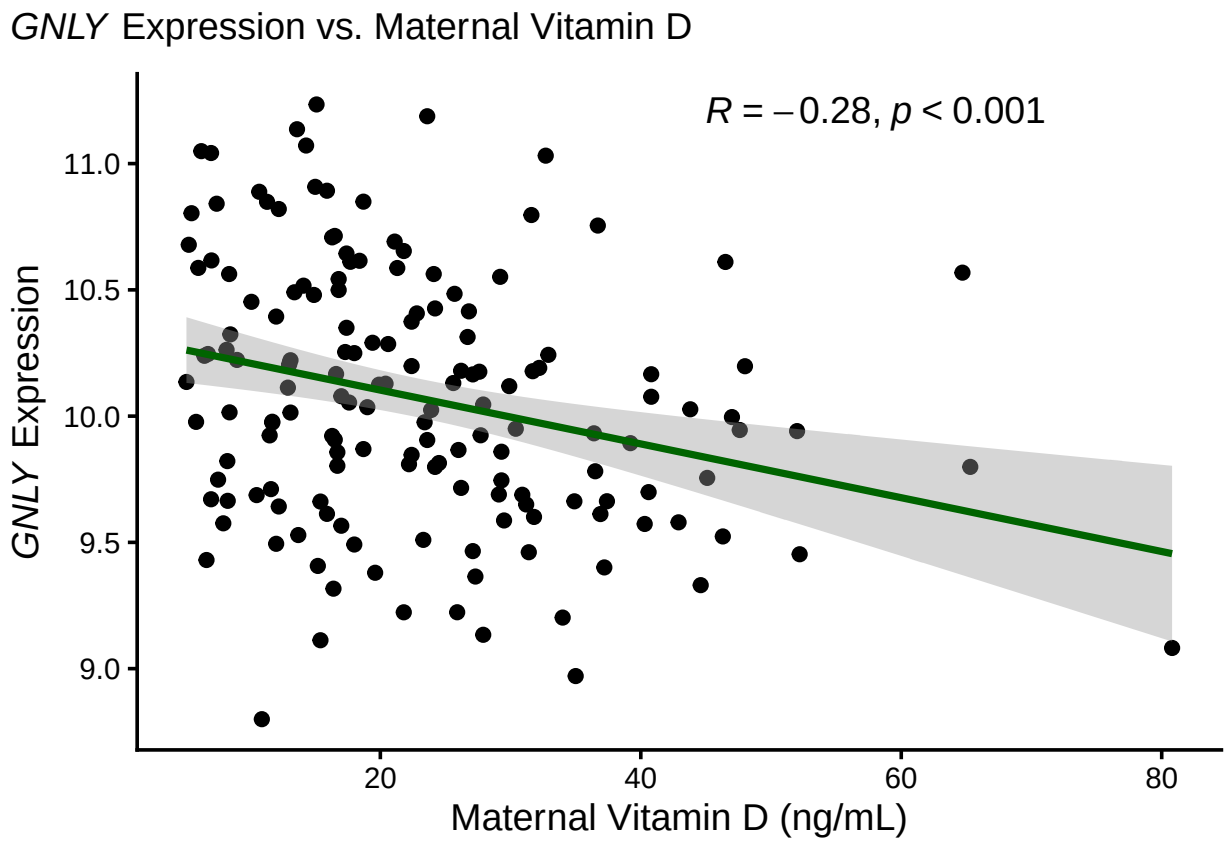

```
ggsave('final_histogram.png',width = 7,height = 5,dpi = 300)
```

0.0.8 Final Plot 2. Continuous covariate vs. gene expression (scatter plot)

```
ggplot(plotData,aes(x=mat_vitD,y=Expression))+
  geom_point(size = 2)+
  geom_smooth(method = lm, color = 'darkgreen')+
  ggpubr::stat_cor(method = 'pearson',
                   label.x = 45,
                   label.y = 11.2,
                   p.accuracy = 0.001)+
  # label.x / label.y move the label to the empty upper-right corner;
  # p.accuracy = 0.001 stops it from printing a long p-value
  labs(title =
         expression(paste(italic("GNLY"),' Expression vs. Maternal Vitamin D')),
    x='Maternal Vitamin D (ng/mL)',
    y=expression(paste(italic('GNLY'),' Expression')))+
```

```
  theme_classic(base_size = 14)+
  theme(plot.title = element_text(size = 14),
                    plot.title.position = 'plot')
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
ggsave('final_scatter.png',width = 7,height = 5,dpi = 300)
```

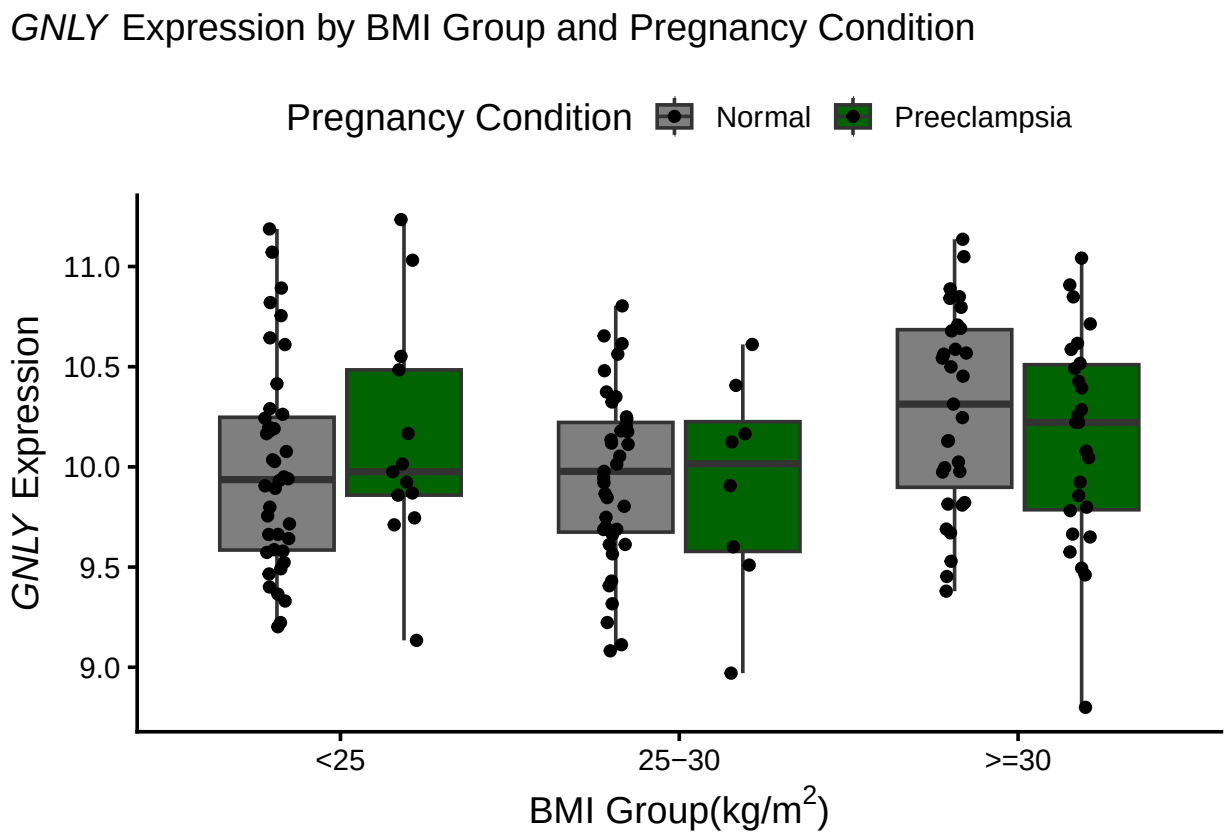
```
## `geom_smooth()` using formula = 'y ~ x'
```

0.0.9 Final Plot 3. Boxplot by both categorical variables

```
# Define the color palette
colorPalette <- c('grey50','darkgreen')

ggplot(plotData,aes(x=bmi_group,y=Expression,fill=preg_preeclampsia))+
  geom_boxplot(outlier.shape = NA)+
  # outlier.shape = NA hides the boxplot's own outlier dots, because every
  # sample is drawn as a point below
  geom_point(aes(group=preg_preeclampsia),
             position = position_jitterdodge(jitter.width = 0.15),
             size = 1.5)+
```

```
  # Individual samples on top of the boxes. It spreads points inside each box
  # so they do not overlap.
  scale_fill_manual(values = colorPalette)+
  labs(title = expression(paste(italic('GNLY'),
                    ' Expression by BMI Group and Pregnancy Condition')),
      x=expression('BMI Group(kg/m'^2*')'),
      y=expression(paste(italic('GNLY'),' Expression')),
      fill='Pregnancy Condition')+
  theme_classic(base_size = 14)+
  theme(legend.position = 'top')+
  theme(plot.title = element_text(size = 14),
                  plot.title.position = 'plot')
```




```
ggsave('final_boxplot.png',width = 7,height = 5,dpi = 300)
```

0.0.10 Heatmap of the 10 most variable genes

```
# Put the expression columns in the same order as the (sorted) covirate rows
expr <- expr[,covar$geoID]

# Calculate variance of each gene
variance <- apply(expr,MARGIN = 1 ,FUN=var)
# Order rows of expr so that highest variance in expression is on top
```

```r
exprOrdered <- expr[order(variance,decreasing = T),]

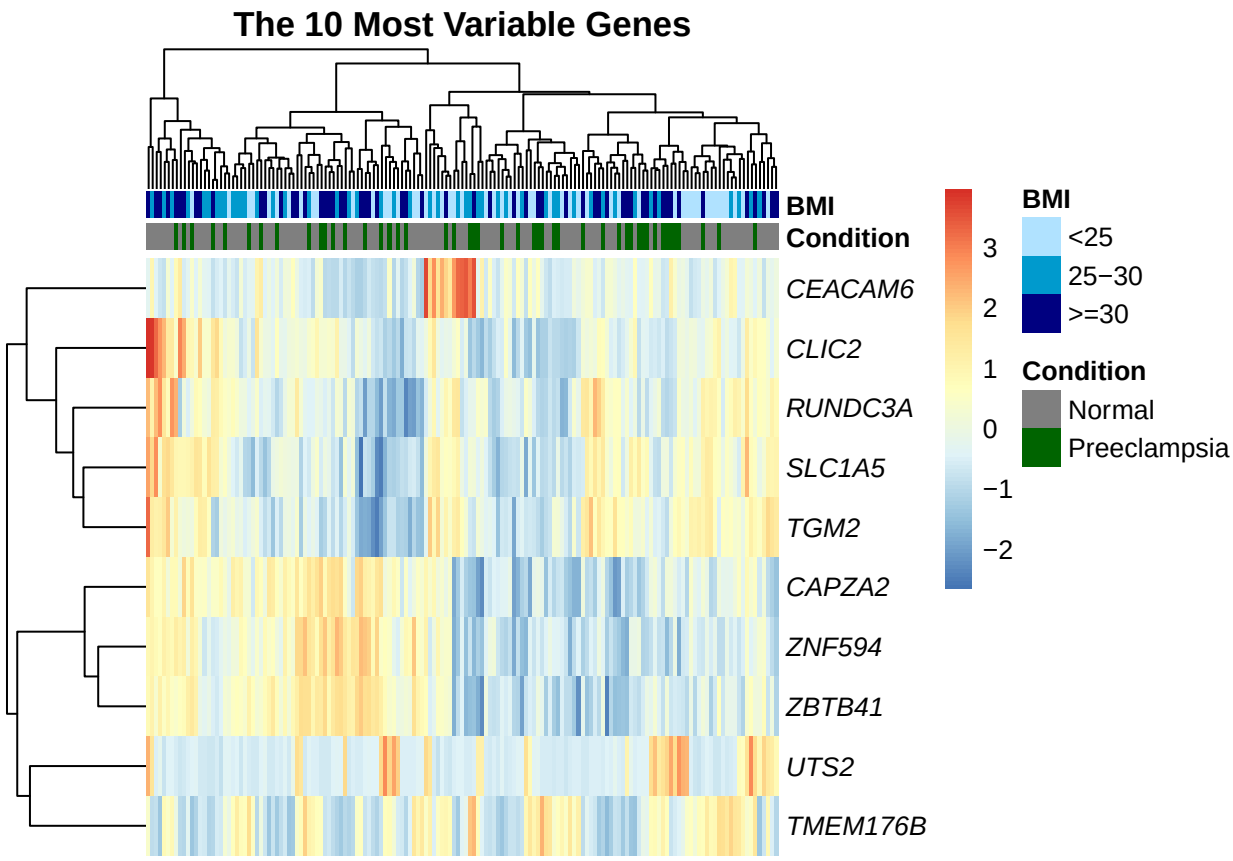
# Keep the top 10 genes
heatData <- exprOrdered[1:10,]

# Change the row names from probe IDs to gene names
row.names(geneNames) <- geneNames$PROBEID
row.names(heatData) <- geneNames[row.names(heatData),'GENE']

# Put every gene on the same scale (mean 0, sd 1 across samples) so that genes
# with high expression do not hide the others
heatData <- t(apply(heatData,MARGIN = 1,FUN = function(x) (x - mean(x))/sd(x)))

# Gene names in italics for the row labels
italicNames <- parse(text = paste0('italic("',row.names(heatData),'")'))

# Define covariates for the 2 traking bars
annotationData <- data.frame(row.names = colnames(heatData),
                             'Condition'=covar$preg_preeclampsia,
                             'BMI'=covar$bmi_group)
annotationColors <- list(Condition=c('Normal'='grey50',
                                     'Preeclampsia'='darkgreen'),
                         BMI=c('<25'='lightskyblue1',
                               '25-30'='deepskyblue3',
                               ' 30'='navy'))
# Generate the heatmap
heatmap <- pheatmap(heatData,
         clustering_distance_cols = 'euclidean',
         clustering_distance_rows = 'euclidean',
         annotation_col = annotationData,
         annotation_colors = annotationColors,
         main = 'The 10 Most Variable Genes',
         labels_row = italicNames,
         # show_colnames=F because 157 sample names would not fit
         show_colnames = F)
```




```
# Save the heatmap (the gtable is the picture inside the pheatmap object)
# Save the heatmap as a png (the png device can draw  , the pdf device could not)
ggsave('final_heatmap.png',plot = heatmap$gtable,width = 8,height = 4.5,
       dpi = 300, bg='white')
```

0.0.11 Additional plot: volcano plot of vitamin D and every gene

A volcano plot shows how big the difference is (x axis) and how strong the evidence is (y axis) for many genes at once. Here very gene is compared between women with vitamin D $\geq$ 30 ng/mL (the usual cut point for sufficient vitamin D) and women with vitamin D $<$ 30 ng/mL.

```
# Define the vitamin D groups
covar$vitD_group <- factor(ifelse(covar$mat_vitD >= 30,' 30','<30'),
                           levels = c('<30',' 30'))

# Define somewhere to store results
volcanoData <- data.frame(row.names = row.names(expr),
                          'Diff' = rep(NA,nrow(expr)),
                          'P.Val' = rep(NA,nrow(expr)))

# Loop through all genes (same idea as the t test function from class)
for (i in row.names(expr)) {
  # Subset data
  covar$Gene <- exprT[covar$geoID,i]
  # Run t test
```

```
  tempTest <- t.test(Gene ~ vitD_group,covar)
  # Extract values: mean of the >=30 group minus mean of the <30 group
  volcanoData[i,'Diff'] <- tempTest$estimate[2] - tempTest$estimate[1]
  volcanoData[i,'P.Val'] <- tempTest$p.value
}

# Add gene names (row names of geneNames were set to the probe IDs above)
volcanoData$Gene <- geneNames[row.names(volcanoData),'GENE']

# Color the genes by direction if p < 0.05
volcanoData$Direction <- ifelse(volcanoData$P.Val < 0.05,
                                ifelse(volcanoData$Diff > 0,
                                       'Higher if  30','Lower if  30'),
                                'p   0.05')

# Bonferroni p value cutoff for 500 tests
bonferroni <- 0.05/nrow(volcanoData)

# The genes I want to name on the plot, and how far above (+) or below (-) its
# point each name goes (GNLY and IL26 have other points right next to them)
labelData <- volcanoData[volcanoData$Gene %in% c('GNLY','LYN','IL26'),]
labelOffset <- c('GNLY' = -0.6,'IL26' = 1.8,'LYN' = 0.5)

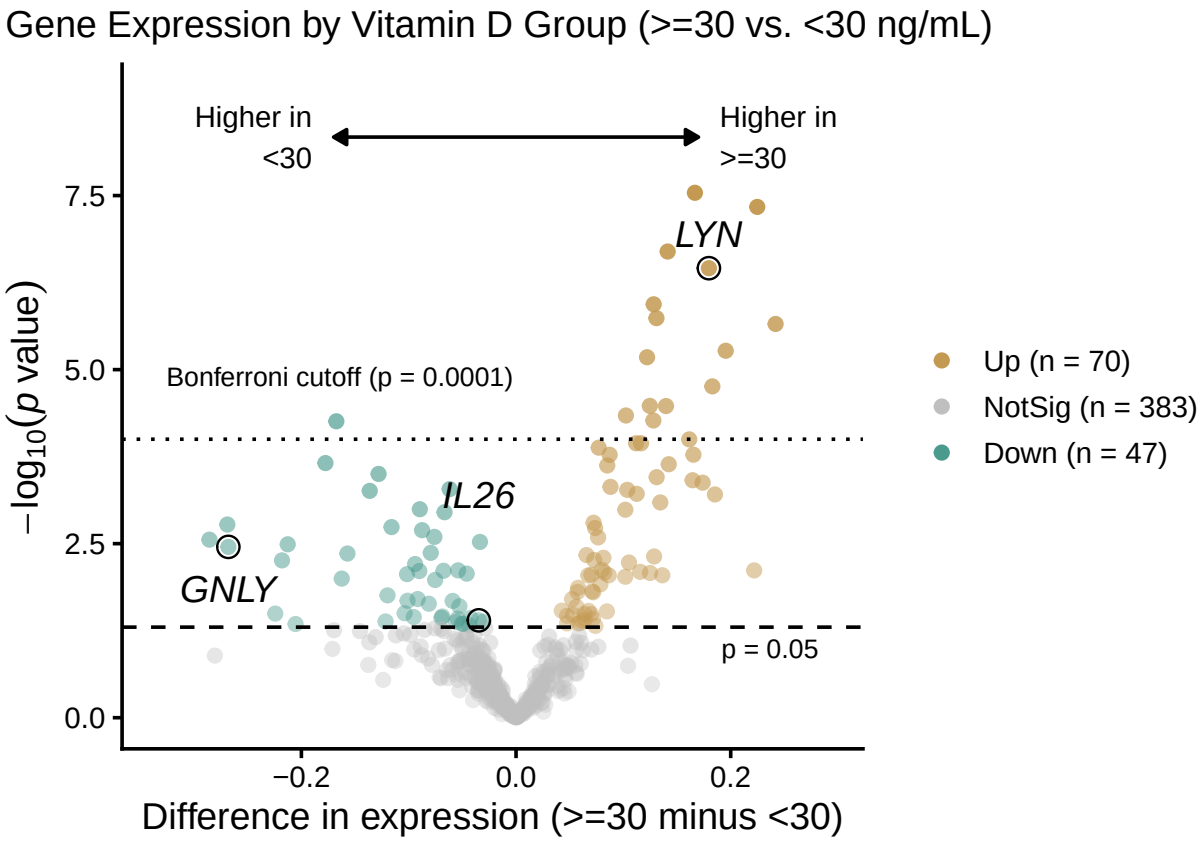
# Muted colors: teal = higher in the <30 group, tan = higher in the  30 group
volcanoColors <- c('Lower if  30' = '#4A9B8E',
                   'Higher if  30' = '#C49A52',
                   'p   0.05' = 'grey75')

# Number of genes in each group (used in the legend)
volcanoCounts <- table(volcanoData$Direction)

# Left and right edges of the plot, and the height of the arrow at the top
xMin <- min(volcanoData$Diff) - 0.05
xMax <- max(volcanoData$Diff) + 0.05
arrowY <- max(-log10(volcanoData$P.Val)) + 0.8

ggplot(volcanoData,aes(x = Diff,y = -log10(P.Val),color = Direction,
                       alpha = -log10(P.Val))) +
  # Add one point per gene
  geom_point(size = 2) +
  # Define colors
    # Legend: Up / NotSig / Down with the number of genes in each group
  scale_color_manual(values = volcanoColors,
                     breaks = c('Higher if  30','p   0.05','Lower if  30'),
                     labels = c(paste0('Up (n = ',volcanoCounts['Higher if  30'],')'),
                                paste0('NotSig (n = ',volcanoCounts['p   0.05'],')'),
                                paste0('Down (n = ',volcanoCounts['Lower if  30'],')'))) +
  # The more significant the gene, the darker its point (no legend for this)
  scale_alpha_continuous(range = c(0.3,1),guide = 'none') +
  # Change scale of the axes to give us more space for the labels
  xlim(xMin,xMax) +
  ylim(0,arrowY + 0.6) +
  # Add a dashed line at p = 0.05 and a dotted line at the Bonferroni cutoff
```

```
annotate(geom = 'segment',x = -Inf,xend = Inf,
         y = -log10(0.05),yend = -log10(0.05),linetype = 'dashed') +
annotate(geom = 'segment',x = -Inf,xend = Inf,
         y = -log10(bonferroni),yend = -log10(bonferroni),linetype = 'dotted') +
# Label the two lines
annotate(geom = 'text',x = xMax - 0.01,y = -log10(0.05) - 0.3,
         label = 'p = 0.05',hjust = 1,size = 3.5) +
annotate(geom = 'text',x = xMin + 0.01,y = -log10(bonferroni) + 0.9,
         label = 'Bonferroni cutoff (p = 0.0001)',hjust = 0,size = 3.5) +
# Add an arrow at the top that says which side is higher in which group
annotate(geom = 'segment',x = -0.17,xend = 0.17,y = arrowY,yend = arrowY,
         arrow = arrow(ends = 'both',type = 'closed',length = unit(0.2,'cm'))) +
annotate(geom = 'text',x = -0.19,y = arrowY,label = 'Higher in\n<30',
         hjust = 1,size = 3.8) +
annotate(geom = 'text',x = 0.19,y = arrowY,label = 'Higher in\n 30',
         hjust = 0,size = 3.8) +
# Circle the 3 genes of interest
geom_point(data = labelData,shape = 21,size = 3.5,color = 'black',fill = NA,
           alpha = 1) +
# Add the names of the 3 genes
# (parse = T turns the text italic(GNLY) into an italic GNLY)
annotate(geom = 'text',x = labelData$Diff,
         y = -log10(labelData$P.Val) + as.numeric(labelOffset[labelData$Gene]),
         label = paste0('italic("',labelData$Gene,'")'),parse = T,
         color = 'black',alpha = 1) +
# Change labels
labs(title = 'Gene Expression by Vitamin D Group ( 30 vs. <30 ng/mL)',
     x = 'Difference in expression ( 30 minus <30)',
     y = expression(-log[10](italic(p)~'value')),color = NULL) +
# Set theme
theme_classic(base_size = 14) +
theme(legend.position = 'right') +
theme(plot.title = element_text(size = 14),
                  plot.title.position = 'plot')
```




```
ggsave('final_volcano.png',width = 7,height = 5,dpi = 300)
```